C. Instruction checks and heuristic index

Archived codes by workflow. Each task row has CANN and CUDA columns; G uses ROCm/HIP in the second column.

Task		M9 Pinning		M10 Steps		M11 Index†	
		CANN	CUDA‡	CANN	CUDA‡	CANN	CUDA‡
Environment and installation							
I	Version compatibility	5	5	5	5	0.73	0.85
J	Installation	4	4	5	5	0.80	0.88
K	Container setup	4	4	5	5	0.86	0.98
Operator development							
C	Library selection	4	5	4	5	0.77	1.00
D	Custom operator	2	5	3	5	0.41	0.88
L	Operator accuracy	3	4	4	4	0.69	0.84
M	Dynamic shapes / tiling	4	4	4	5	0.63	0.94
N	Operator fusion	4	4	4	4	0.74	0.83
O	Framework integration	4	4	5	5	0.72	0.84
Training							
F	Distributed training	3	4	3	5	0.72	0.88
P	Mixed precision	4	5	4	5	0.83	0.98
Q	Memory / OOM	3	4	4	5	0.86	0.94
R	Training convergence	3	4	4	4	0.75	0.98
Inference and deployment							
A	Model conversion	3	5	4	5	0.75	0.94
H	Quantization	4	4	5	5	0.69	0.88
S	Inference serving	4	4	4	5	0.74	0.94
T	Dynamic inference	4	5	4	5	0.72	0.92
Performance and optimization							
B	Profiling	3	4	3	5	0.70	0.93
U	Memory / occupancy	4	4	4	4	0.88	1.00
V	Compute-transfer overlap	4	5	4	5	0.72	0.98
W	Multi-device communication	4	5	4	5	0.70	0.92
Debugging							
E	Error-code diagnosis	3	4	3	5	0.55	0.99
X	Memory-error diagnosis	4	4	5	5	0.74	0.88
Migration							
Y	Chip migration	4	5	4	5	0.68	0.92
Z	Programming concepts	2	2	2	2	0.90	0.92
Migration analogy (not in CANN/CUDA summaries)							
G	Migration analogy [CANN / ROCm-	4	4	4	5	0.84	0.88

M1–M10 are archived ordinal codes (1–5; higher is more favorable; M8 denotes lower effort). "—" = unobserved core adequacy.

* M7 is an archived estimate. † M11 is the historical heuristic availability index. ‡ For G, CUDA‡ = ROCm/HIP; it is excluded from CANN/CUDA summaries.